

L2 Processes Threads

1. What is a process in an operating system?

- a) A static entity stored on disk
- b) A running program with an address space
- c) A thread of control within a program
- d) A collection of open files

Answer: b

2. Which of the following is NOT part of a process's memory layout?

- a) Code
- b) Stack
- c) Heap
- d) Cache

Answer: d

3. What does the `fork()` system call do?

- a) Creates a new thread
- b) Creates a new process by cloning the parent process
- c) Terminates a process
- d) Suspends the execution of a process

Answer: b

4. Which system call replaces the current process image with a new one?

- a) `fork()`
- b) `wait()`
- c) `exec()`
- d) `exit()`

Answer: c

5. What is the purpose of the `wait()` system call?

- a) To create a new process
- b) To wait for I/O operations to complete
- c) To suspend the parent process until a child process terminates
- d) To destroy a process

Answer: c

6. What is stored in the Process Control Block (PCB)?

- a) Process ID, state, and parent pointer
- b) Only the stack pointer and program counter
- c) The entire code and data of the process
- d) Kernel threads only

Answer: a

7. In which state is a process when it is waiting for an I/O operation to complete?

- a) READY
- b) RUNNING
- c) BLOCKED
- d) TERMINATED

Answer: c

8. What does the `execvp()` function do?

- a) Creates a new thread in user space
- b) Executes a program specified by its path and arguments array
- c) Waits for child processes to terminate
- d) Suspends a thread

Answer: b

9. What is the difference between `wait()` and `waitpid()`?

- a) `waitpid()` allows specifying which child process to wait for, while `wait()` does not.
- b) `wait()` waits for all processes, while `waitpid()` waits only for threads.
- c) Both are identical in functionality.
- d) `waitpid()` suspends processes, while `wait()` terminates them.

Answer: a

10. What is the primary advantage of threads over processes?

- a) Threads have separate address spaces.
- b) Threads are cheaper to create and manage than processes.
- c) Threads cannot share resources like processes can.
- d) Threads only exist in kernel mode.

Answer: b

11. What does multithreading allow in modern operating systems?

- a) Multiple address spaces per thread

- b) Concurrent execution within the same address space
- c) Execution of only one thread at any time in a system
- d) Elimination of kernel threads

Answer: b

12. Which type of thread is managed entirely by user-level libraries?

- a) Kernel threads
- b) User-level threads
- c) System threads
- d) Lightweight kernel threads

Answer: b

13. Which API function creates a new thread in POSIX systems?

- a) ``pthread_exit()``
- b) ``pthread_create()``
- c) ``pthread_wait()``
- d) ``pthread_signal()``

Answer: b

15. In user-level threading, what manages thread scheduling?

- a) Operating system kernel
- b) User-level thread library
- c) Hardware interrupts
- d) System calls

Answer: b

16. Which of these is NOT true about kernel threads?

- a) They are managed by the operating system kernel.
- b) They are more expensive than user-level threads for fine-grained tasks.
- c) They require no context switching overhead.
- d) They allow overlapping I/O and computation inside a process.

Answer: c

17. Why are user-level threads faster than kernel threads?

- a) They bypass kernel involvement for common operations like creation and synchronization.
- b) They have dedicated CPU cores assigned to them directly by hardware.

- c) They do not require stack memory allocation during creation.
- d) They avoid all forms of context switching overheads entirely.

Answer: a

18. What does the term "process tree" refer to?

- a) A hierarchical representation of all running threads within an OS
- b) A structure showing parent-child relationships among processes
- c) A list of all processes sorted by their priority levels
- d) A binary tree used for scheduling algorithms

Answer: b

19. What happens during a thread context switch at user level?

- a) Kernel saves and restores hardware states like PC and SP using system calls.
- b) Machine state is saved on one thread's stack and restored from another's stack by user code.
- c) All open files are closed before switching contexts between two threads in the same process.
- d) The OS allocates additional memory for each thread switch dynamically during runtime.

Answer: b

20. Which command can be used to display all processes as a flat list in Linux?

- a) ``pstree``
- b) ``ps``
- c) ``top``
- d) ``lsproc``

Answer: b